Grade 8
Unit 11
Lesson 7 Practice

Name Answer Key
Date
Period

Késhaun conducted an experiment where he dropped a ball from different heights and measured the height of the rebound using feet (ft). He conducted the experiment three times. His data is shown.

Height of Ball Drop (ft)		3	4.5	5	6.5	7	12
Height of Rebound (ft)	Trial 1	2.1	3.7	3.9	5.4	5.6	10.1
	Trial 2	2.7	3.6	4	5	5.3	9.3
	Trial 3	2.4	3.5	4.3	5.2	5.8	9.7

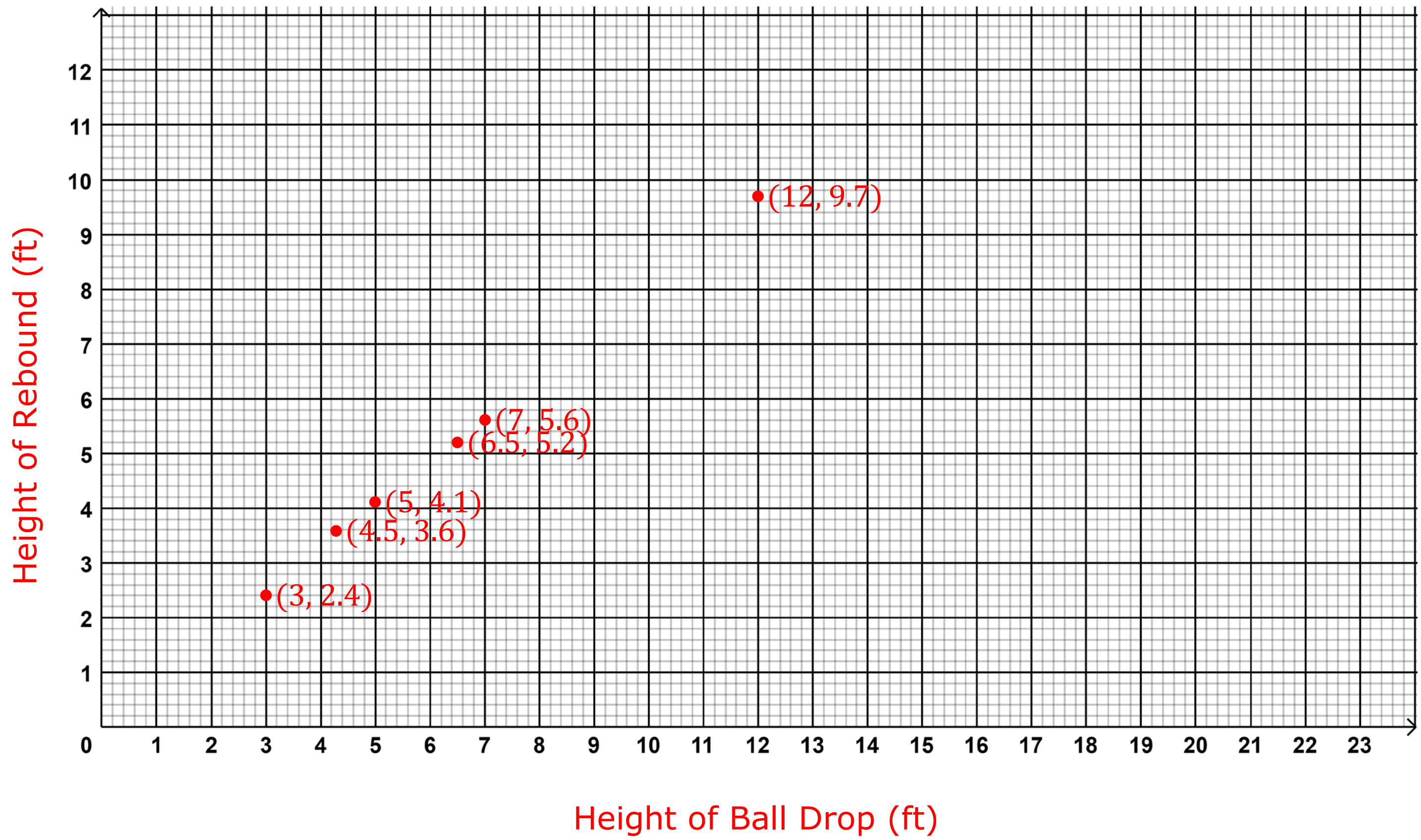
Consider the quantities being compared in this experiment.

1. Identify the independent variable for the experiment.
   Height of Ball Drop
2. Identify the dependent variable for the experiment.
   Height of Rebound

Find the average height, in ft, for the three trials. Complete the table.

#		
3.	**Height of Ball Drop (ft)**	3
	Average Height of Rebound (ft)	$\frac{2.1+2.7+2.4}{3}=\frac{7.2}{3}=2.4$
4.	**Height of Ball Drop (ft)**	4.5
	Average Height of Rebound (ft)	$\frac{3.7+3.6+3.5}{3}=\frac{10.8}{3}=3.6$
5.	**Height of Ball Drop (ft)**	5
	Average Height of Rebound (ft)	$\frac{3.9+4+4.3}{3}=\frac{12.2}{3}\approx 4.1$
6.	**Height of Ball Drop (ft)**	6.5
	Average Height of Rebound (ft)	$\frac{5.4+5+5.2}{3}=\frac{15.6}{3}=5.2$
7.	**Height of Ball Drop (ft)**	7
	Average Height of Rebound (ft)	$\frac{5.6+5.3+5.8}{3}=\frac{16.7}{3}\approx 5.6$
8.	**Height of Ball Drop (ft)**	12
	Average Height of Rebound (ft)	$\frac{10.1+9.3+9.7}{3}=\frac{29.1}{3}=9.7$

9. Label the independent and dependent variables on the appropriate axes.



10. Using the average height of the ball rebound, plot the ordered pairs that represent the variables in the experiment.